

UBCO Freshwater Science (B.Sc.) → Professional Designations (BC)

Purpose: Quick checklist to help students map the Freshwater Science major to common BC designations and pick electives strategically. Always confirm current rules with the regulator.

A) Registered Professional Biologist (RPBio) — College of Applied Biologists (CAB)

- Core academics: Bachelor's degree + 25 science courses (13 in biology) + core courses: communications; chemistry; numeracy; statistics at 2nd-year or higher; applied biology/management; ecology; and 3 of 5 from genetics, cell biology, physiology, systematics/taxonomy, evolution.
- Freshwater fit: Communication (first-year writing); Chemistry (CHEM 121/123, 201/210/211); Numeracy (MATH 100/101); Statistics (STAT 230 or GEOG 271 or BIOL 202 at 2nd-year+); Ecology/applied biology (BIOL 201, BIOL 308, FWSC/BIOL 375).
- Electives to close gaps: BIOL 200 (Cell), BIOL 265 (Genetics), one of BIOL 356/358 (Physiology) or BIOL 371 (Systematics).
- Experience path: BIT → RPBio with validated experience, competencies, and work products.

B) Professional Agrologist (P.Ag.) — BC Institute of Agrologists (BCIA)

- Core academics: Bachelor's degree with 8 foundational 100/200-level science courses + 20 agrology-related courses (including 8 senior at 3rd/4th-year).
- Freshwater fit: Year 1 science + EESC 205/222 cover foundations; upper-level EESC/BIOL/CHEM/GEOG/GISC/ECON can satisfy the 20 agrology courses with ≥8 senior.
- Electives (examples): EESC 301/309/313/314/315/333/342/402/413/422/423/429/431/435; GEOG/GISC 380/381; ECON 371/372; CHEM 301/302/317.
- Experience path: AAg → P.Ag. after articling/competencies.

C) Professional Geoscientist (P.Geo.) — EGBC (Environmental Geoscience)

- Evaluation: EGBC self-evaluation checklist against foundations + compulsory geoscience + environmental areas.
- Freshwater strengths: MATH 100/101; CHEM 121/123; PHYS 111/112; Hydrology (EESC 205); Geochemistry (EESC 309/323/423); Geomorph/Soils (EESC 222). Choose Hydrogeology (EESC 342) and GIS/RS (GEOG/GISC 380/381).
- Add to satisfy compulsory geoscience: Mineralogy/Petrology (EESC 200 or 201), Sed/Strat (EESC 356), Structural (EESC 325), Field Techniques (EESC 435). Recommended: Geophysics (EESC 350).
- Experience path: GIT → P.Geo.; exams may be assigned if gaps remain.

D) Professional Engineer (P.Eng.) — EGBC

- Reality check: P.Eng. normally requires a CEAB-accredited engineering degree. A B.Sc. alone typically doesn't meet standard academics.
- Options: EIT with assigned academics (case-by-case), second degree/MEng/bridging, or P.L.Eng. after specialized experience.
- Helpful electives: ENGR 341/342 (seat/prereq limits apply), APSC 254, GISC 380/381, EESC 413, programming/data (STAT/DATA).

Sample 4■Year Study Plans (with Co■op Options)

Advising note: Illustrative only. Verify current UBCO Calendar, term offerings, prerequisites, and seat restrictions. “ex.” marks examples; substitutes with comparable content may also work for your regulator.

Track 1 — RPBio (College of Applied Biologists)

- Year 1 — T1: BIOL 116; CHEM 121/123 (one); MATH 100; WRDS/ENGL/COMM; elective.
- Year 1 — T2: BIOL 125; CHEM 123/121 (the other); PHYS 111 or 112; elective.
- Year 2 — T1: BIOL 201; STAT 230 or GEOG 271 or BIOL 202 (ensure ≥2nd■year stats); EESC 205; elective.
- Year 2 — T2: EESC 222; CHEM 211; BIOL 200 (Cell); elective.
- Year 3 — T1: EESC 301 (Limnology); FWSC/BIOL 375; CHEM 301; elective.
- Year 3 — T2: BIOL 265 (Genetics); BIOL 356/358 (Physiology) or BIOL 371 (Systematics); BIO upper■level.
- Year 4 — T1: EESC 402; EESC 309/323/423 (choose 1–2); BIO upper■level (reach 13 BIO courses).
- Year 4 — T2: EESC 342 or EESC 413; BIO upper■level; Directed Studies/Field/Special Topics.
- Co■op: after Year■2 (summer start) for 8–12 months or Year■3 summer. Target biomonitoring, fisheries, consulting, government.
- Seat/prereq cautions: some 3xx BIO seats prioritize Biology majors; request permissions early.

Track 2 — P.Ag. (BC Institute of Agrologists)

- Year 1 — T1: BIOL 116; CHEM 121/123; MATH 100; WRDS/ENGL; elective.
- Year 1 — T2: BIOL 125; CHEM 123/121; PHYS 111/112; elective.
- Year 2 — T1: EESC 205; STAT 230 / GEOG 271 / BIOL 202; ECON 371/372 (ex.); elective.
- Year 2 — T2: EESC 222; CHEM 211; CHEM 201/210 (ex.); elective.
- Year 3 — T1: EESC 301; EESC 402 or other policy/management upper■level (ex.); GISC/GEOG 380/381 (GIS/RS).
- Year 3 — T2: Choose among EESC 333/413/422/429/431/435 (ex.); consider CHEM 301/302 as relevant.
- Year 4: Round out to ≥20 agrology■related courses with ≥8 at 3rd/4th■year across EESC/BIOL/CHEM/GEOG/GISC/ECON; add field/industry projects.
- Co■op: after Year■2 or Year■3; target utilities, local/regional governments, consulting, agriculture/watershed NGOs.
- Seat/prereq cautions: GIS may have priority rules; ECON seats fill quickly; plan early.

Track 3 — P.Geo. (Environmental Geoscience, EGBC)

- Year 1 — T1: CHEM 121/123; MATH 100; PHYS 111/112; BIOL 116; elective.
- Year 1 — T2: CHEM 123/121; MATH 101/103 (ex.); PHYS 112/121 (ex.); BIOL 125; elective.
- Year 2 — T1: EESC 205; EESC 200 or 201 (Mineralogy/Petrology — compulsory); STAT 230 or GEOG 271; elective.
- Year 2 — T2: EESC 222; Intro GIS (ex. GEOG/GISC 272/280 if offered) or defer to 3rd year; CHEM 211; elective.
- Year 3 — T1: EESC 301; EESC 325 (Structural — compulsory); EESC 309/323 (Geochemistry); GISC/GEOG 380/381 (GIS/RS).
- Year 3 — T2: EESC 356 (Sed/Strat — compulsory); EESC 350 (Geophysics — recommended); upper■level elective.

- Year 4 — T1: EESC 342 (Hydrogeology); EESC 423 (Advanced Geochem, ex.); elective.
- Year 4 — T2: EESC 435 (Field Techniques — compulsory/field); capstone/Directed Studies aligned to environmental geoscience.
- Co-op: after Year 2 into Year 3; target environmental consulting, mining environment, groundwater, government hydrology.
- Seat/prereq cautions: Some EESC 3xx/4xx need specific lower division geoscience; take EESC 200/201 early. GIS/RS courses have priority rules.

Disclaimer: This advising pack is illustrative. Verify all requirements with current CAB, BCIA, and EGBC rules and the UBCO Academic Calendar. Course availability and seat priorities vary by term.